

Review Questions: Factorial, Split-Plot, Strip-Plot, and Linear Mixed Models

Exam 2 Preparation — AGST 50104 Experimental Design, Spring 2026

AGST 50104 — Spring 2026

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Question 1

A 3×3 factorial experiment (Factor A = herbicide product, 3 levels; Factor B = application timing, 3 levels) is conducted in a CRD with 5 replications. The interaction $A \times B$ is statistically significant ($p < 0.001$). What is the most appropriate next step?

- a) Report the main-effect means for herbicide and timing first, then mention the interaction as a secondary finding.
 - b) Refit the study as a one-way ANOVA on the nine treatment combinations and ignore the factor structure.
 - c) Examine the interaction effects because the effect of herbicide depends on timing and cannot be summarized by one overall main effect.
 - d) Conclude that neither factor has a stable effect, so no further treatment comparisons are justified.
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Question 2

In a $2 \times 2 \times 2$ factorial experiment (Factors A, B, and C), the three-way interaction $A \times B \times C$ is significant. What does this mean in plain language?

- a) Each main effect is significant once the other two factors are included in the model.
 - b) Factor C changes the overall mean response but does not affect the $A \times B$ relationship.
 - c) The $A \times C$ and $B \times C$ interactions offset each other when the data are averaged over factor B.
 - d) The $A \times B$ interaction differs across levels of C, so it must be interpreted separately within each level of C.
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Question 3

A 3×3 factorial experiment (Factor A: 3 levels; Factor B: 3 levels) is run in a CRD with 8 complete replications. How many degrees of freedom does the $A \times B$ interaction have?

- a) 4
 - b) 2
 - c) 8
 - d) 63
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Question 4

The **identity** covariance structure is the simplest residual variance model in a linear mixed model. Which statement correctly describes the identity structure?

- a) All pairs of residuals from the same group share a common non-zero correlation and a common variance.
 - b) Each group or time-point has its own residual variance, but residuals remain uncorrelated.
 - c) Every observation has the same residual variance and all residual covariances are zero.
 - d) Residual variances and covariances are all estimated freely with no structural restrictions.
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Question 5

A weed scientist studies the interaction of tillage system (no-till vs. conventional) and POST herbicide rate (0, 280, 560 g ai/ha) on Palmer amaranth density. Tillage is applied to large field sections; herbicide rate is applied to small subplots within each section. Which statement correctly identifies the design?

- a) This is a split-plot design with tillage as the whole-plot factor and herbicide rate as the subplot factor.
 - b) This is a strip-plot design because both factors are imposed by field equipment rather than by hand.
 - c) This is an RCBD factorial because every treatment combination appears within each block.
 - d) This is a CRD because herbicide rate, not tillage, is randomized to the smallest units.
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Question 6

A split-plot field trial has 4 complete blocks. Irrigation system (2 levels) is assigned to whole plots, and nitrogen rate (3 levels) is randomized to subplots within each whole plot. Which statement correctly matches the error stratum used to test each fixed effect?

- a) Irrigation is tested against subplot error with 12 df, and nitrogen is tested against whole-plot error with 3 df.
 - b) Both irrigation and nitrogen are tested against one pooled residual because every treatment combination appears in every block.
 - c) Irrigation is tested against whole-plot error with 6 df, and nitrogen is tested against subplot error with 12 df.
 - d) Irrigation is tested against whole-plot error with 3 df, and nitrogen is tested against subplot error with 12 df.
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Question 7

In each of 4 blocks, a weed scientist randomly assigns cover crop species to plots in a row direction, next they independently assign POST herbicide timing to each plot in the column direction. Weed biomass is measured at the intersections of the two randomization systems. Which option best identifies both the design and the analysis strategy?

- a) Split-plot design: `lmer(y ~ cover * timing + (1 | block) + (1 | block:cover), data = d)`
 - b) Strip-plot design: `lmer(y ~ cover * timing + (1 | block) + (1 | block:cover) + (1 | block:timing), data = d)`
 - c) RCBD factorial: `lm(y ~ block + cover * timing, data = d)`
 - d) Latin square: `aov(y ~ block + row + col + cover + timing, data = d)`
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Question 8

A weed scientist analyzes a split-plot study with `lm(density ~ block + tillage * herbicide, data = d)` even though tillage was assigned to whole plots and herbicide to subplots. What is the most likely consequence?

- a) Herbicide will be tested against an error term that is too large, so subplot effects lose power.
- b) The tillage $\times$ herbicide interaction cannot be estimated because split-plot layouts are not factorial.
- c) The tillage effect will be tested against an error term that is too small because whole-plot variation was ignored.
- d) Including block as a fixed effect automatically accounts for the split-plot error structure.

Question 9

Which feature most clearly distinguishes a **strip-plot** design from a **split-plot** design?

- a) Both factors are applied in crossing strips, creating separate error strata for each main effect and for their interaction.
 - b) One factor is assigned to whole plots and the second factor is randomized to subplots nested within them.
 - c) Every treatment combination is randomized independently to small plots within each block.
 - d) The design confounds treatments with rows and columns so that only one factor can be estimated.
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Question 10

A weed scientist compares 4 specific herbicide programs across 10 randomly selected farms over 3 growing seasons. The goal is to compare those exact 4 programs, while drawing inference beyond the observed farms and beyond those particular years. How should these factors be treated in the model?

- a) Herbicide program, farm, and year should all be fixed because they were all observed explicitly.
 - b) Herbicide program should be fixed, while farm and year should be random.
 - c) Herbicide program should be random because it is one source of variation, while farm and year should be fixed.
 - d) Herbicide program and farm should be fixed, while year should be random because weather cannot be controlled.
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Question 11

A plant pathologist tests three fungicide products at five research stations. The **same three products** are applied at **all five stations**. Which statement is correct?

- a) Fungicide products are nested within station; a different set of products is used at each station.
 - b) Fungicide products and stations are crossed because every product appears at every station, allowing a product $\times$ station interaction to be estimated.
 - c) Station is nested within fungicide product because each station applies only one product.
 - d) The design is confounded and the main effects of neither factor can be estimated.
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Question 12

In an on-farm trial, three technicians (**tech**) are each assigned to a different farm (**farm**), so each technician works at only one farm. Technician effects need to be in the random structure. Which `lmer()` notation correctly expresses that technician is **nested within** farm?

- a) `(1 | tech:farm)` alone, because the nested term already absorbs the farm-to-farm variation
 - b) `(1 | tech) + (1 | farm)`, because technicians and farms are crossed random effects
 - c) `(1 | farm) + (1 | farm:tech)` or equivalently `(1 | farm/tech)`, because technicians are unique within farms
 - d) `(1 | tech * farm)`, because the `*` operator tells `lmer()` that technician is nested within farm
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Question 13

A researcher fits two nested mixed models for yield data from a multi-location trial:

- Reduced: `m0 <- lmer(yield ~ variety + (1 | location), data = d, REML = FALSE)`
- Full: `m1 <- lmer(yield ~ variety + fungicide + (1 | location), data = d, REML = FALSE)`

They want to test whether adding fungicide improves model fit using a likelihood ratio test. Which statement is correct?

- a) Compare `m0` and `m1` with a likelihood ratio test because the models are nested and both were fit by ML.
 - b) Refit both models with `REML = TRUE` before using the likelihood ratio test because REML is always preferred.
 - c) A likelihood ratio test is inappropriate here because fungicide is a fixed effect rather than a random effect.
 - d) The test is only valid if both models have exactly the same number of parameters.
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Question 14

A crop scientist runs a multi-location trial of 4 herbicide-resistant soybean varieties across 8 randomly selected farms in the region. She intends to generalize findings beyond these specific farms. How should varieties and farms be classified in the model?

- a) Varieties and farms are both fixed because the observed levels are completely known.
- b) Varieties and farms are both random because the model includes more than one source of variation.

- c) Varieties are random and farms are fixed because farms are the physical experimental units.
 - d) Varieties are fixed and farms are random because the varieties were chosen deliberately but the farms represent a broader population.
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Question 15

A researcher fits `lmer(yield ~ variety + (1 | location), data = d)` and finds the estimated location variance is essentially zero (near 0.001). What is the most appropriate interpretation?

- a) The model is invalid because random-effect variances cannot be close to zero in practice.
 - b) Location contributes very little variability after fixed effects are included, so the random location term adds little to the analysis.
 - c) The fixed effects are necessarily non-significant because the location term absorbed all treatment variation.
 - d) The data must contain a measurement problem because an estimated variance near zero is not interpretable.
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Question 16

When estimating variance components in a linear mixed model, why is **REML** generally preferred over full **ML**?

- a) REML is preferred mainly because most software cannot fit mixed models by ML.
 - b) REML and ML give the same variance estimates once the fixed effects are specified correctly.
 - c) REML adjusts for the estimation of fixed effects before estimating variances, so variance components are less biased in finite samples.
 - d) REML must be used whenever a model contains more than one random effect.
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Question 17

A researcher measures weed density on the same plot at 0, 2, 4, and 6 weeks after treatment and fits `lmer(density ~ treatment * week + (1 | plot), data = d)`. What does `(1 | plot)` do in this model?

- a) It gives each plot its own baseline level, accounting for correlation among repeated measurements from the same plot.
- b) It allows the treatment effect to have a different slope over time for every plot.

- c) It forces the residual variance to be different at each sampling week.
 - d) It guarantees that measurements within a plot are independent once week is included.
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Question 18

A researcher fits three competing models to the same dataset and obtains the following:

Model	AIC	BIC
M1	-380	-390
M2	-408	-421
M3	-315	-324

Which model is preferred if the priority is balancing **goodness of fit and parsimony**?

- a) M1, because the largest AIC signals the strongest model fit.
 - b) M2, because it has the lowest AIC and the lowest BIC among the candidate models.
 - c) M3, because values close to the minimum are always preferred when they come from a simpler model.
 - d) None, because information criteria should not be used to compare models fit to the same response.
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Question 19

A multi-environment maize trial is analysed in ASReml-R. Two residual specifications are compared using information criteria:

- Model A: `residual = ~ id(units)` (identity)
- Model B: `residual = ~ dsum(~ id(units) | env)` (diagonal)

What is the key structural difference between these two specifications?

- a) Model A gives each environment its own residual variance, whereas Model B forces one common residual variance.
 - b) Model B adds genotype-by-environment interaction as a fixed effect, whereas Model A leaves it out.
 - c) The two specifications are equivalent because `id(units)` is evaluated separately within environment by default.
 - d) Model A uses one common residual variance for all environments, whereas Model B allows a different residual variance within each environment.
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Question 20

The **diagonal** covariance structure differs from the **identity** structure in what specific way?

- a) The diagonal structure allows non-zero correlations within group, whereas identity assumes zero correlation.
- b) The diagonal structure has fewer parameters than identity because it removes all variance terms.
- c) The diagonal structure keeps residuals uncorrelated but allows variances to differ across groups or time points, whereas identity also requires equal variances.
- d) The diagonal structure is only for repeated measures, whereas identity is only for multi-environment trials.

Answer Key

Q1 — Answer: c

When the interaction is significant, the effect of Factor A depends on the level of Factor B (and vice versa). Reporting only main-effect averages in that situation is misleading because the average “washes out” differences that exist among the simple effects. The correct strategy is to examine simple effects. Recasting the study as a one-way ANOVA does not solve the interpretation problem; it merely hides the factorial structure that created it.

Q2 — Answer: d

A significant three-way interaction means that the two-way $A \times B$ relationship is not constant across levels of C. In practice, you must examine the $A \times B$ interaction separately for each level of C. This is the direct extension of how a two-way interaction invalidates simple main-effect comparisons.

Q3 — Answer: a

For a two-factor factorial, interaction $df = (a - 1)(b - 1) = (3 - 1)(3 - 1) = 2 \times 2 = 4$. The number of replications does not affect the interaction degrees of freedom; it affects only the error degrees of freedom.

Q4 — Answer: c

The identity structure assumes each observation has the same residual variance (σ^2) and that all residuals are uncorrelated. This is the standard ordinary ANOVA assumption: errors are independent and identically distributed. Option a describes compound symmetry; option b describes a diagonal structure with unequal variances; option d describes an unstructured matrix.

Q5 — Answer: a

The defining criterion for a whole-plot factor is not scientific importance but **practical randomization constraint**. Tillage requires equipment that covers large areas; it cannot be independently assigned to small individual plots. Herbicide rate can be applied plot-by-plot with a sprayer, making it the subplot factor.

Q6 — Answer: d

In a split-plot, the whole-plot factor is tested against whole-plot error, not subplot error. With 4 blocks and 2 whole-plot levels, whole-plot error $df = (4 - 1)(2 - 1) = \mathbf{3}$. The subplot factor is tested against subplot error, with $df = 2 \times (4 - 1) \times (3 - 1) = \mathbf{12}$. Recognizing which error stratum belongs to which factor is more important than memorizing the formulas separately.

Q7 — Answer: b

Because the two treatment systems are imposed as perpendicular strips, the design is a **strip-plot**, not a split-plot. The correct analysis keeps separate error strata for the cover-crop strips, the herbicide-timing strips, and their intersection plots within each block. A one-error-term RCBD analysis would ignore the defining feature of the layout.

Q8 — Answer: c

Ignoring the whole-plot random variation makes the tillage factor look as if it had been randomized to all subplots independently. That uses an error term that is too small for testing tillage, which inflates the F-statistic and makes the whole-plot factor appear more significant than it should.

Q9 — Answer: a

In a split-plot, one factor (WPF) is applied to whole plots; the other (SPF) is applied within each whole plot. There are **two** error strata: whole-plot error and subplot error. In a strip-plot, both factors are applied to strips running perpendicular to each other. This crossing generates **three** error strata — one for each main factor and one for their interaction — each tested against a different error term.

Q10 — Answer: b

The herbicide programs are **fixed** because the researcher wants inference on those exact 4 named programs. Farms and years are **random** because they represent samples from larger populations of possible farms and seasons, and the goal is to generalize beyond the observed set. Treating farms or years as fixed would narrow the inference to only those particular units.

Q11 — Answer: b

When all levels of Factor A appear with all levels of Factor B, the factors are **crossed**. Here, all three fungicide products are present at all five stations — a full product $\times$ station factorial structure. Nesting would require that each station use a *different set* of products.

Q12 — Answer: c

Nesting is written in `lmer()` as `(1 | A/B)` or equivalently `(1 | A) + (1 | A:B)`. This tells the model that the levels of B (technician) are unique to each level of A (farm). Option a omits the farm main effect, and option b implies crossed random effects — every technician appears at every farm — which contradicts the scenario.

Q13 — Answer: a

The likelihood ratio test compares nested models by their log-likelihoods, so the reduced and full models must be fit with **ML** when the fixed-effect structures differ. Here, `m0` is nested within `m1`, and both were fit with `REML = FALSE`, so `anova(m0, m1)` provides the appropriate likelihood ratio test. Requiring REML here would be incorrect because REML likelihoods are not directly comparable across models with different fixed effects.

Q14 — Answer: d

Variety is **fixed** because the researcher deliberately selected those four specific varieties to compare directly — the same four varieties would be used in a follow-up study and named in the conclusions. Farm is **random** because the eight farms are a sample from all possible farms in the region; the researcher wants to generalize beyond these eight specific farms.

Q15 — Answer: b

A near-zero random effect variance tells you that the grouping variable (location) contributes negligible variability once fixed effects are accounted for. The intraclass correlation would be close to zero, meaning observations within the same location are no more similar than those from different locations. The random term may be retained for design-correctness, but it changes the analysis very little.

Q16 — Answer: c

Maximum likelihood (ML) treats fixed-effect parameters as known when estimating variance components, so it does not account for the uncertainty (degrees of freedom) used up by fixed effects. REML marginalizes over fixed effects before estimating variance components, producing unbiased estimates — especially important in small samples with many fixed-effect parameters.

Q17 — Answer: a

The random intercept (`1 | plot`) gives each plot its own baseline level, which accounts for the fact that repeated observations from the same plot are more alike than observations from different plots. It does not create random slopes, does not impose heterogeneous residual variances, and does not make within-plot observations independent.

Q18 — Answer: b

AIC and BIC both penalize model complexity. Lower values indicate a better trade-off between fit and parsimony. M2 has the **lowest AIC (-408) and the lowest BIC (-421)**, making it the preferred model under both criteria. When AIC and BIC agree, the choice is straightforward.

Q19 — Answer: d

`residual = ~ id(units)` specifies the identity structure: one scalar residual variance applies to all observations. `residual = ~ dsum(~ id(units) | env)` specifies the diagonal structure: a separate residual variance is estimated within each environment, allowing fields to differ in measurement precision. Both still assume independence across environments. AIC or BIC (fitted with ML, not REML) can be used to decide whether the extra parameters in the diagonal model are warranted.

Q20 — Answer: c

The identity structure constrains all residual variances to be equal (one σ^2) and all off-diagonal covariances to be zero. The diagonal structure relaxes only the equal-variance constraint: each group or time-point gets its own σ^2_k , but all off-diagonal elements remain zero. This makes the diagonal structure appropriate when groups are known to differ in their measurement precision (heteroscedastic, independent errors) — for example, environments in a multi-environment trial where some fields are more variable than others.